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Beyond the Job Posting: What Hiring Managers Seek in Entry-Level Software Engineering Candidates

Spencer Baloga Loufek
Cloud Platform Infrastructure
Hewlett Packard Enterprise
Fort Collins, CO, USA
spencer.baloga-loufek@hpe.com

Fabio Santos
Computer Science
Colorado State University
Fort Collins, CO, USA
fabio.deabreusantos@colostate.edu

Bianca Trinkenreich
Computer Science
Colorado State University
Fort Collins, CO, USA
bianca.trinkenreich@colostate.edu

Abstract*—Background: With limited job openings and growing selectivity in tech, understanding the skills hiring managers expect from entry-level Software Engineering candidates is essential. While job postings emphasize technical abilities, success often hinges on non-technical skills, as interpersonal and adaptive abilities. Yet, mismatches between candidate profiles and employer expectations lead to inefficient hiring and missed opportunities. Aims: We investigated which skills hiring managers expect from early-career Software Engineering (SE) candidates and how those skills are described in job postings. Method: To achieve this goal, we conducted an exploratory case study in Hewlett Packard Enterprise (HPE), a global technology company. We interviewed 12 managers, collected seven HPE job postings related to entry-level positions in Software Engineering-related roles, and employed mixed methods to analyze the data. Results: Our findings show that hiring managers place strong emphasis on non-technical expectations when evaluating entry-level candidates. They commonly believe that technical competencies can be taught on the job, provided the candidate demonstrates the right non-technical attributes. The non-technical expectations included adaptability, proactivity, dependability, self-direction, passion, collaboration, communication, problem solving, courage, self-awareness, self-confidence, willingness to learn, and having a process-oriented mindset. Conclusions: To secure their first job, Software Engineering students should invest not only in technical competencies but also in developing non-technical skills. Our findings show that hiring managers prioritize these skills, even though they are often underrepresented in job postings. Making such expectations more visible could help bridge the gap between candidates and employers, leading to more efficient and effective hiring.

***Index Terms*—job, skills, technical, non-technical, job posting**

I. Lay Abstract

Landing a first job in tech can be difficult, especially for recent graduates trying to understand what hiring managers are really looking for. This study investigates which skills are most valued by hiring managers when selecting entry-level candidates for Software Engineering-related roles. We conducted interviews with 12 technical managers at Hewlett Packard Enterprise (HPE), a large global technology company, and analyzed job postings to understand how required skills are communicated during the hiring process. Although job ads often list technical qualifications such as programming languages or tools, managers emphasized that these are not the most important factors when evaluating candidates. They consistently reported that non-technical skills, such as communication, adaptability, collaboration, and a willingness to learn, play a critical role in hiring decisions. Technical skills are viewed as teachable, while non-technical abilities are seen as more difficult to develop after hiring. However, these non-technical expectations are often missing or vague in job postings. This creates confusion for candidates who may over-focus on technical preparation and underprepare for the interpersonal and behavioral aspects of interviews. Hiring managers, in turn, struggle to identify well-rounded applicants from resumes alone. Improving how these skills are communicated in job descriptions and interviews could help reduce mismatches, improve hiring outcomes, and make the recruitment process more transparent. By better aligning expectations and preparation, this study contributes to more effective early-career hiring in the software industry.

II. Introduction

In a rapidly evolving tech industry, technical proficiency alone no longer guarantees success for entry-level Computer Science candidates [1]. While job postings from leading companies still list programming languages and cloud platforms, top tech employers are increasingly recognizing the importance of non-technical skills such as adaptability, communication, and cross-functional collaboration. According to Gartner's industry-wide analysis [2], progressive organizations are not just tracking shifts in technical expertise; they are also investing in broader learning and development strategies to cultivate agile, well-rounded talent. This trend reflects a growing consensus that enduring success in technology roles requires the ability to learn continuously, navigate ambiguity, and contribute effectively to diverse teams.

In software engineering, early-career hiring often suffers from a disconnect between the skills valued by employers and those emphasized in job postings or academic training. While prior studies highlight the importance of non-technical skills for software roles [1, 3], these competencies are frequently underrepresented in recruitment materials. This gap complicates hiring and preparation for industry needs.

Our study addresses this challenge through a case study at Hewlett Packard Enterprise (HPE), a multi-cultural and large

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global software development organization that has more than 60,000 employees around the world, revealing how hiring managers prioritize skills (most of those, non-technical) that are not clearly articulated in job ads.

Therefore, our goal is to identify the skills that hiring managers expect from entry-level candidates for Software Engineering(SE)-related jobs and how those skills are presented in job postings. To achieve our goal, we defined the following research questions:

RQ1. *What do hiring managers expect from candidates for entry-level Software Engineering-related positions?*

RQ2. *How are these expected skills reflected in job postings for entry-level Software Engineering-related positions?*

To answer our research questions, we collected data through 12 interviews with HPE managers about their expectations regarding the candidates for entry-level SE-related job positions, and collected seven different job postings, also from HPE, for entry-level Software Engineering-related positions. We conducted a qualitative analysis of the interviews and identified 13 non-technical and five technical skills expected by HPE hiring managers. We then enriched each skill category with its corresponding description and used large language models (LLMs), through a Retrieval-Augmented Generation (RAG) approach, to assess whether these expected skills were present in the job postings collected from HPE.

Our results show that hiring managers at HPE prioritize non-technical skills over specific technical proficiencies when evaluating entry-level SE candidates. However, these skills are often underrepresented or vaguely described in job postings. From the best of our knowledge, this is the first study to directly compare SE managers expectations and job artifacts in a structured way within a global company.

These findings have practical implications for industry which we discuss in Sec. V. Improving alignment between hiring expectations and job descriptions can support more effective, skills-based recruitment. Educators can also use these insights to better prepare students for real-world hiring. This paper contributes to the empirical understanding of early-career hiring by analyzing both manager expectations and job artifacts. It highlights actionable gaps and offers recommendations to improve recruitment practices in software engineering.

III. Research Design

To answer our research questions, we conducted an exploratory case study [4] via interviews with HPE managers from various business units that hire entry-level candidates for SE-related jobs, including Hybrid Cloud, Application-Specific Integrated Circuit Design, Compute, High Performance Computing & Artificial Intelligence. In this section, we describe the case and the phases of planning, data collection, and analysis [5].

A. The Case and Unit of Analysis

Our case is a software development organization within Hewlett Packard Enterprise (HPE), a global tech company with over 60,000 employees. HPE emphasizes both technical and non-technical skills in hiring and supports career growth through its Technical Career Path (TCP) program. As the company shifts toward skills-based hiring, it aims to assess candidates by demonstrated competencies rather than credentials. However, misalignment between job postings and actual expectations remains a challenge, as gaps between stated and actual hiring criteria lead to misaligned candidate expectations.

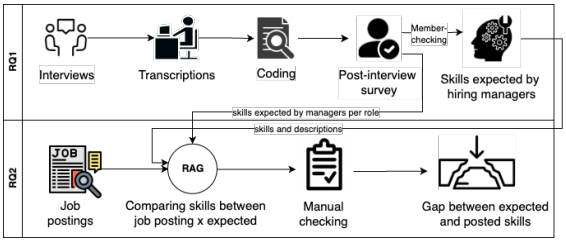


Fig. 1: Research design

Our study examines this issue as a contemporary phenomenon within its real-life context [6], using a case study of HPE's software development organization to analyze the gap between formal job descriptions and the competencies managers value during early-career hiring. The findings aim to inform improvements in recruitment communication and support HPE's ongoing transition to skills-based hiring. This case study was collaboratively conducted by one researcher from the company (the first author) and two outsiders (the second and third authors). Fig. 1 has our research design.

B. Interview Planning

For the interviews, we targeted HPE managers from technical business units who are directly involved in recruiting early-career candidates for software engineering–related roles within their teams.

We conducted semi-structured interviews [7]. We used a script to guide the different areas of inquiry [8], while also listening for unanticipated information during the flow of the conversation. The interviews revolved around the central question: *"What skills do you value and look for when evaluating candidates for entry-level positions, and how do you assess those skills during the hiring process?"* We approached this topic after establishing rapport with the interviewee, and asking them to describe, without mentioning any names, the perfect and worst candidate they can remember interviewing for an entry-level position.

We interviewed participants until we were unable to find any new expected skills for three consecutive interviews. Our final sample comprised 12 participants as we present in Table I.

C. Data Collection

Interviews: The first author works at HPE and conducted the interviews on Microsoft Teams [1], which lasted between 40 and 60 minutes. With participant consent, we recorded all interviews. The same researcher transcribed the interviews using Microsoft Teams' Transcript automatic feature and listened to each

[1]https://www.microsoft.com/en-us/microsoft-teams/group-chat-software

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TABLE I: Demographics of respondents (n=12)

Part. ID	Gender	HPE Tenure	HPE Role
P1	Man	3 years	Senior Manager
P2	Man	19 years	Director
P3	Man	25 years	Senior Manager
P4	Man	8 years	Senior Manager
P5	Man	23 years	Senior Manager
P6	Woman	1 year	Manager
P7	Man	3 years	Manager
P8	Woman	32 years	Director
P9	Man	7 years	Senior Manager
P10	Man	25 years	Senior Manager
P11	Woman	4 years	Senior Manager
P12	Man	24 years	Senior Director

recording, adjusting the corresponding transcriptions, mainly with respect to technical terms and project names. The project was approved by HPE, and only the first author of the paper had access to de-identified data. Our sample consisted of 12 hiring managers. This is in line with the anthropology literature, which mentions that a set of 10-20 knowledgeable people is sufficient to uncover and understand the core categories in any cultural domain or study of lived experience [9]. Table I presents the demographics of our sample.

Job postings: The first author collected seven job postings [8] from different business units from HPE, that are entry-level, require a Computer Science degree, and the roles are related to Software Engineering.

D. Data Analysis

Interviews: We analyzed the responses to the open questions about the skills that hiring managers expect from candidates for entry-level Software Engineering-related positions. We qualitatively analyzed the answers to open questions by inductively applying open coding [10] to organize what the participants reported. Two authors collaboratively conducted multiple card sorting sessions [11]. Through an iterative process of continuous comparison [12], they refined a set of post-formed codes of skills grounded in the perspectives of the participants (P1–P12). Categorization decisions were made through a negotiated agreement [13], where both researchers discussed the rationale for the placement of each code until a consensus was reached. A third author, who has a high degree of experience in qualitative analysis studies, reviewed the results. The outcome was a set of higher-level categories as cataloged in our codebook in Table III.

Job postings: To systematically find the skills expected by hiring managers with the skills listed in job posting documents, we used Retrieval-Augmented Generation (RAG) with category names and descriptions of skills from RQ1. Indeed, the managers' vision about the skills is used to retrieve relevant data when querying the skills present in Job Postings. This strategy aligns with the principle of providing contextual information as outlined by Geroimenko [14], which emphasizes that supplying relevant background helps guide the model towards more accurate, nuanced, and audience-appropriate responses. In addition, RAG manages to generate, factually accurate responses avoiding hallucinations [15]. We formulated a structured prompt specifically designed to query the presence of the skills (as the result of our RQ1 analysis IV-A) in each job posting. Despite the small number of job postings related to Software Engineering and entry-level positions, RAG option was operated to provide future growth in scale. The code was done in Google Colab [2] using the GPT model openai:o3-mini. The o3-mini model was in production since January 2025 and is the most balanced in terms of cost and efficiency, addresses the technical domains requiring precision and speed, and is optimized for STEM reasoning. The model has no temperature parameter. The only available hyperparameter is reasoningEffort and we set it with medium reasoning to balance speed and level [3]. Three authors manually checked the results against the seven job postings.

E. Post-Interview Survey

To member-checking or qualitative coding of expected skills from interviews, we conducted a follow-up survey with the 12 hiring managers who had agreed to participate further. The survey presented the full list of skill categories and descriptions derived from our RQ1 analysis. Participants were asked to confirm the relevance of each skill, suggest edits to descriptions, or add missing skills. Five managers responded. All confirmed the skill list and descriptions as accurate, and no additional skills were suggested. This feedback helped to strengthen the credibility of our findings by confirming that the coded categories reflected the original intent of the participants. The same survey also supported our analysis for RQ2 by asking participants to review seven entry-level job postings and select the roles they were familiar with. For each selected role, they reviewed a pre-populated list of expected skills (based on the interview analysis) and were invited to confirm, refine, or expand the list through open-text fields. Four managers provided input on Cloud Developer I and Systems/SW Engineer I; two on SW Designer I; and one each on SW Engineer Firmware I, Product Manager I, Test Engineer I, and SW Quality Assurance I. Their feedback informed our comparison between stated expectations and the skill content found in job postings. The full questionnaire is available in the replication package [8].

IV. Analyses and Results

We now address the research questions introduced in Sec. II.

A. RQ. What do hiring managers expect from candidates for entry-level Software Engineering-related positions?

We found five categories of technical skills expectations, and 13 categories of non-technical skills expectations, as presented in Fig. 2. In the following, we present more details about our findings organized by skill category.

1) Technical Skills: Our participants reported to expect knowledge on Cloud Computing, DevOps, Open Source Software, Operating Systems, and Version Control Systems, as we present in Table II.

[2]https://colab.research.google.com/

[3]https://openai.com/index/openai-o3-mini/

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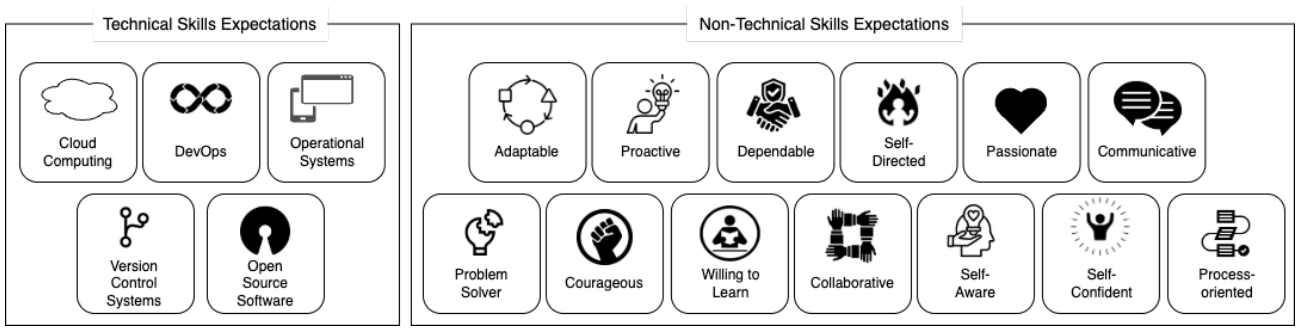


Fig. 2: The expectations reported by HPE hiring managers who participated in our study.

TABLE II: Codebook of Technical Skills Expected from Early-Career CS Candidates

Category Name	Description	Example Quotes
Cloud Computing	Ability to design, deploy, and understand systems on cloud platforms such as AWS, GCP, or Azure, including knowledge of cloud-first architectures and service models.	P1: *"cloud first methodology"*; P1: *"how to design a service that runs in the cloud"*; P1: *"have some experience with go and Terraform doing things in Kubernetes and AWS"*
DevOps	Understanding of infrastructure-as-code tools, continuous integration/continuous deployment (CI/CD), and the cultural and technical practices needed to support cloud infrastructure operations.	P10: *"[recent graduates] don't teach DevOps concepts at school"*; P10: *"modernize some of their computer science curriculum... like cloud infrastructure classes"*; P6: *"I would go to the AWS documentation. I would check out this… I know this community… I could go and find a sample project…"*
Operational Systems	Familiarity with Linux, command-line tools, and OS-level tasks such as terminal navigation and shell scripting. Also includes being truthful about stated OS competencies.	P9: *"they don't have kind of basic Linux skills anymore and that becomes a little bit problematic"*; P9: *"the method of interaction is through a terminal… it is integral to everything we do"*; P3: *"If you say like I'm a Linux bash guy and I start asking you bash questions and you don't know anything about bash, then you probably shouldn't have put it on your resume."*; P8: *"They don't do well at an operating systems"*
Version Control Systems	Knowledge of Git and version control practices for managing code collaboratively, including branching, auditing, and maintaining a single source of truth.	P12: *"version control management practices on source social curation… the craft of software engineering that's not really related to the algorithms… but how you get around in the actual enterprise"*; P6: *"They're learning git, right? So they understand. You know what git is? And you know how we use it for, you know, account auditing and… single source of knowledge for our code…"*
Open Source Software	Hands-on participation in community-led software projects, contributing code and engaging in collaborative development processes.	P12: *"Anybody could go to a [CNCF] architecture meeting and go pick some Kube[rnates] bugs and you'll get the whole experience then, right... what it takes to actually get a patch into a real software project — that's significant."*; P1: *"have some experience with go and Terraform doing things in Kubernetes and AWS"*

Cloud Computing: Hiring managers emphasized the importance of cloud computing skills, particularly for positions in environments driven by modern infrastructure and service delivery models. P1 highlighted the expectation of a *"cloud first methodology"* and the need for candidates to demonstrate familiarity with platforms like AWS, GCP, or Azure, showing awareness of *"the cloud mentality"* and *"how to design a service that runs in the cloud."* These expectations underscore a shift in foundational software engineering practices towards scalable, distributed architectures. Additionally, P1 noted the relevance of hands-on experience with cloud-native technologies, such as *"go and Terraform doing things in Kubernetes and AWS."* As P11 observed, *"most of [the candidates] don't really have cloud experience—they're not being taught cloud classes,"* raising concerns about how well curricula prepare students for real-world technical environments.

DevOps: Several managers expressed concern about the disconnect between current academic curricula and the operational practices expected in modern engineering teams. P10 pointed out that *"[instructors] usually don't teach DevOps concepts at school"* and recommended that institutions *"modernize some of their computer science curriculum... like cloud infrastructure classes."* To bridge this educational gap, managers value candidates who take initiative and seek out learning resources independently. For instance, P6 described their own learning approach: *"I would go to the AWS documentation. I would check out this… I know this community… I could go and find a sample project…"* These insights suggest that familiarity with DevOps tools and workflows, such as CI/CD pipelines and infrastructure-as-code, is seen as essential for readiness in production environments.

Open Source Software: Hiring managers at HPE expressed that open source software (OSS) experience can meaningfully differentiate entry-level candidates by demonstrating not only technical competence but also the collaborative mindset needed in real-world engineering environments. P12 emphasized the value of hands-on participation in open ecosystems, stating, *"Anybody could go to a [CNCF] architecture meeting and go pick some Kube[rnates] bugs and you'll get the whole experience then, right... what it takes to actually get a patch*

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into a real software project, that's significant." For this manager, OSS involvement acts as a proxy for readiness to work with large-scale systems, highlighting a candidate's ability to contribute to community-led development cycles and engage in practices like peer reviews, bug triaging, and CI/CD workflows. P1 reinforced this expectation through their emphasis on practical cloud-native toolchains, noting that ideal candidates should *"have some experience with go and Terraform doing things in Kubernetes and AWS."*

Operating systems knowledge, particularly related to Linux and terminal interaction, emerged as a technical expectation among hiring managers. P9 noted a concerning trend in recent graduates, stating that *"they don't have kind of basic Linux skills anymore and that becomes a little bit problematic."* This gap is especially critical given that, in their context, *"the method of interaction is through a terminal... it is integral to everything [HPE] does."* Beyond basic usage, managers also expect candidates to have a truthful and functional grasp of the tools they claim to know. As P3 pointed out, *"If you say like I'm a Linux bash guy and I start asking you bash questions and you don't know anything about bash, then you probably shouldn't have put it on your resume."* In some cases, weaknesses in this area surface during technical assessments, as P8 observed: *"They don't do well at an operating systems."*

Version control systems, particularly Git, were identified by hiring managers as a foundational expected skill. P12 emphasized that understanding *"version control management practices on source social curation"* is part of the broader *"craft of software engineering that's not really related to the algorithms... but how you get around in the actual enterprise."* This perspective highlights the importance of knowing how to collaborate effectively in real-world codebases, where navigating branches, resolving merge conflicts, and maintaining semantic versioning are critical to team productivity and software stability. Similarly, P6 acknowledged that most new graduates today *"are learning git, right?"* and pointed to the practical importance of version control in the workplace, explaining that *"you know how we use it for, you know, account auditing and... single source of knowledge for our code..."*

2) Non-Technical Skills: Our participants mentioned to expect the candidate to be ADAPTABLE, PROACTIVE, DEPENDABLE, SELF-DIRECTED, PASSIONATE, COLLABORATIVE, EFFECTIVE COMMUNICATOR, PROBLEM SOLVER, COURAGEOUS, SELF-AWARENESS, SELF-CONFIDENT, WILLING TO LEARN, and PROCESS-ORIENTED MINDSET (Table III).

Adaptable: Our participants emphasized that early career SE candidates must be able to adapt to unexpected problems and changes (P5, P9). Not just adapt, but *"adapt quickly"* (P9), and then *"pivot and engage"* (P5). This is essential because, as P9 explained, *"tools and methodologies change very frequently, so being adaptable and flexible is really important"*. P5 illustrated a scenario where adaptability is critical: *"You're executing the strategy. That's undefined, and so that first part of it is developing a strategy and scoping the change that you need to make, and then figuring out the systems and measures that need to be in place in order to adapt that change."*

Proactive: Our participants expect early-career workers to demonstrate taking *"initiative"* (P2, P4), which is *"hard to find during an interview"*. P3 explained this initiative as being a *"self starter"* and illustrated it with a real situation: *"I'm going to be able to give them an assignment and then I'm not going to have to observe them. I'm going to check in with them, but they're going to go and figure out how to get started."* Since the question focused on early-career workers, P3 clarified they were *"not suggesting to just throw them in the deep end of the pool"* but rather emphasized the importance of *"getting them peers that they can reference"* and *"giving them a work item, they're going to pick up that work item"*. P3 further explained that initiative is not just about being assigned a task, but about actively seeking one.

Self-directed: Managers expect a candidate to have self-motivation (P1,P11,P12), not just the extrinsic motivation, but the *"intrinsic motivation, or what gets them up in the morning"* (P2). Besides taking the initiative to work on a task, having also initiative to learn (P5,P9). P11 mentioned as *"a person who pushes himself"* on attitude or initiative that can be related to learn or work on any activity at work. P9 described an example of initiative as learning a new tool by running it after reading the documentation, and remembered about a candidate that had the initiative to dive on a tool being used by HPE, so the candidate was *"demonstrating that even though [they] didn't know about this big thing that [HPE] uses, they went and learned about it. And [HPE] ultimately hired that person."*

Dependable was mentioned as someone managers and team members can *"trust"* (P2,P3). P3 highlighted the importance of trust as even higher priority than being an outstanding achiever, by saying that *"if you get someone that is an outstanding achiever, but nobody trusts them"* it would be a bad choice, and gave an example of a situation of lack of trust when someone tries *"to steal credit for the work that [other people]*.

Passionate: According to the employee work passion theory, work passion arises when the employee repeatedly interpret their work environment positively, leading them to feel good emotionally, find their work meaningful, and wanting to act in ways that help the organization (e.g., performing well, staying, endorsing it) [16]. P7 emphasized the importance of having passion as he would *"lower a little bit the requirements and focus more on their passions."* As P9 stated: *"I want to have a candidate work on my team who is when they wake up in the morning, they're excited to come to work and tackle exciting challenges that we're trying to solve. I don't want someone to wake up in the morning and be like, oh, I got to go to work again. That's not a fun person to work with."*

Collaborative is a characteristic that includes being a team player and willing to work with others [17]. Managers expect entry-level candidates to demonstrate the ability to engage in team environments, with several calling teamwork a key hiring criterion. P8 emphasized, *"teamwork is my number one thing"* and *"teamwork is the deal breaker."* P1 reinforced its importance, explaining that *"working with a team... is when we get the best solutions"* due to diverse perspectives, yet noted that many candidates lack this experience because *"so much of*

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TABLE III: Codebook of Non-Technical Skills Expected from Early-Career CS Candidates

Category name	Description	Example Quotes
Adaptable	Adjusts quickly to changes and uncertain situations.	"adapt quickly" (P9), "pivot and engage" (P5), "tools and methodologies change very frequently" (P9)
Proactive	Takes initiative to start tasks and solve problems without being prompted.	"initiative" (P2, P4), "self starter" (P3), "not just about being assigned a task, but actively seeking one" (P3)
Self-directed	Motivated to act and learn independently.	"intrinsic motivation" (P2), "went and learned about it" (P9)
Dependable	Trusted by others and behaves with integrity.	"trust" (P2, P3), "if you get someone that is an outstanding achiever, but nobody trusts them" (P3)
Passionate	Emotionally engaged with their work and motivated by purpose.	"focus more into their passions" (P7), "excited to come to work" (P9)
Collaborative	Works well in teams and values diverse contributions.	"teamwork is the deal breaker" (P8), "no lone wolf work here" (P9)
Communicative	Listens, speaks clearly, and adapts communication to the audience.	"listen and asked good critical probing questions" (P2), "read the audience" (P3), "describe things from a technical nature" (P10)
Problem Solver	Breaks down and analyzes problems creatively and collaboratively.	"problem solving mentality" (P1), "partitioning complex problems" (P7), "how do you approach a problem with no solution" (P5)
Courageous	Faces uncertainty, asks questions, and admits limits without fear.	"courage to be vulnerable" (P5), "takes courage to ask questions and be wrong" (P10)
Self-Aware	Understands personal strengths, weaknesses, and emotional responses.	"ability of self regulation" (P2), "know your limitations" (P3), "difference between urgency and importance" (P12)
Self-confident	Comfortable sharing ideas and handling ambiguity.	"self-confidence on trying things out" (P2), "comfortable communicating their ideas" (P3)
Willing to Learn	Eager to learn and improve continuously, driven by curiosity.	"always be learning" (P2), "I don't know that, but I'm willing to learn" (P6), "learnability, ability to climb the curve" (P8)
Process-Oriented	Understands how their work fits into larger systems and optimizes for efficiency.	"process thinking" (P6), "how does this affect everything else around it" (P6), "intentional with releases" (P12)

the emphasis is on doing your own work." Still, P1 saw it as *"a skill that people can learn,"* and focused interview questions on past team projects. P3 observed candidates' interaction with teams to assess how well they *"mesh with the team,"* while P9 clearly stated, *"there is no lone wolf work here."* Although P7 admitted collaboration *"is very, very hard,"* it remains critical, especially in customer-facing and cross-functional projects. As P12 summarized, managers need to know *"how do they behave in a team or do they thrive in the team environment,"* even if these behaviors are hard to assess.

Communicative candidates, or those who listen actively, express ideas clearly, and adapt to their audience, are highly valued by the managers in our sample. P2 emphasized emotional intelligence, noting strong candidates *"listened and asked good critical probing questions"* and showed *"a degree of clarity"* in their responses. P3 stressed that effective communicators must also *"read the audience"* to gauge whether their message is landing. For software engineering roles, the ability to clearly explain technical concepts is especially critical. As P5 put it, candidates must be *"able to articulate something technical or able to convey an idea or an architecture in simple terms that can be consumed."* P10 echoed this need, explaining that it is *"very valuable in our teams as software engineers because at some point, they're going to do a presentation on their project, a design review, and you have to be able to read the room and describe things from a technical nature in a way that your audience can understand it."*

Problem Solvers are expected to not only apply technical knowledge but also demonstrate adaptability and analytical thinking when tackling problems. Managers emphasized that candidates must show a *"problem solving mentality"* (P1), an ability to *"solve problems with a team"*, *"know how to solve problems with code."* Several managers described problem solving as both a mindset and a structured process. P7 valued candidates who could *"break down a problem in a micro algorithmic way"*, *"partitioning complex problems into small problems."* P10 similarly explained that good problem solvers *"break it down in your own brain and then apply some computer science concept to be able to solve it."* In practical team settings, P7 noted that early-career developers are often *"trying to solve a real life problem for a particular business,"* where context and collaboration matter. P5 stressed the importance of process, asking candidates *"how do [they] approach a problem for which there is no solution"* and whether they can *"convey a technical problem in a way that somebody can engage and work with [them]."* Resilience and flexibility also matter. P9 pointed out that candidates should know how to handle *"unexpected problems, whether a bug in their code or there was a global pandemic."*

Courageous candidates, those willing to take risks, admit what they don't know, and engage openly, are highly valued by managers hiring early-career engineers. P5 highlighted that *"courage to be vulnerable"* is a key differentiator, noting that early-career professionals often struggle because *"they're not sure of themselves or they are oversure of themselves. And in both scenarios, the outcome results in hiding, citing things you don't know."* Managers emphasized that true growth in technical roles requires the bravery to confront uncertainty and failure. As P5 put it, *"you don't get good at something hard without courage."* This is particularly relevant in fast-paced engineering environments where learning is continuous and errors are part of the process. P10 echoed this, noting that junior developers need *"the courage to ask questions that might expose you to something that you need to learn,"* and that *"it takes courage to ask questions and then be wrong."* Together, these insights show that courage is not just about boldness but

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about embracing the discomfort of learning and improving in technical and collaborative settings.

Self-Aware emerged as a critical expectation from managers evaluating early-career candidates, particularly for roles that demand both technical competence and interpersonal adaptability. P2 explained that self-awareness includes *"an ability of self regulation,"* especially under pressure, observing how candidates respond emotionally when *"the pressure is on."* Emotional intelligence was frequently mentioned as a related trait. P2 described it as the ability to *"step outside yourself to look at the situation with a different set of eyes,"* and to seek feedback and advice. This capacity for reflection and growth is what P2 called a sign of *"good emotional maturity,"* emphasizing that while perfection isn't expected, *"you can't be oblivious either."* P4 reinforced this idea, describing self-awareness as understanding both oneself and others: *"it's about them understanding themselves and those around them."* P3 similarly stressed the importance of recognizing personal limits, stating, *"you have to know your limitations."* P6 added a nuanced dimension, focusing on resilience in the face of uncertainty: *"the gap between I don't know something and I know something,"* and whether candidates can *"sit in that in between and feel the frustration"* without giving up. P10 even used candidates' demeanor in interviews to assess fit, noting that if someone *"can't relax... they're probably 'wired pretty tight,'"* which may or may not align with the team's needs when working under pressure for a software delivery. Finally, P12 summed it up as *"understanding yourself and how you'll best work,"* including recognizing priorities such as the *"difference between urgency and importance."*

Self-Confident: Managers highlighted self-confidence as a critical attribute for entry-level candidates, often linking it to the ability to navigate ambiguity and take initiative. P3 described self-confidence as being *"comfortable communicating their ideas,"* while P2 emphasized the importance of having *"self-confidence on trying things out and not being afraid to fail,"* noting that early-career workers are often *"more nervous of looking bad and fail."* This hesitation, P2 explained, relates to *"a whole psychological aspect, but when someone feels good, they have a healthy EQ."* Reflecting on a memorable candidate, P2 shared that *"what made her unique in those cases was she would follow up, saying that because [she] didn't know, here's what [she] would do to go find this."* This proactive mindset, grounded in confidence, stood out even more as the candidate *"came across as somebody who was a solid thinker, brought critical thinking skills and just had a confidence and calmness,"* which P2 found *"really impressive for somebody of her age,"* and more typical of someone *"from an executive in mid 50s."*

Willing to Learn was consistently emphasized by managers as an essential trait for early-career candidates entering software and technical roles. P1 captured this expectation simply: *"people who have learned and want to continue learning as they go."* Managers explained that strong candidates show a proactive, curious mindset, what P2 called a commitment to *"developing a growth mindset"* and the practice of *"always be learning, always be looking for new opportunities to learn,"* including beyond just technical topics into interpersonal and collaborative skills. Several highlighted that willingness to learn is especially important when navigating the unknown. As P3 noted, candidates who say, *"I don't know something, but I'm willing to go learn it,"* often stand out, especially when paired with the self-awareness and initiative to *"figure out on their own the steps [they] have to take to be a self-starter."* This aligns with P6's view that for junior hires, *"the biggest thing you have... is 'I'm here to learn'"* and that a simple statement like *"I don't know that, but I'm willing to learn"* is often enough to earn a manager's confidence. Curiosity was also considered a core signal for willingness to learn. P7 asked himself during an interview, *"does this candidate have curiosity? Is this candidate inquisitive?"* and celebrated a past hire who *"wasn't afraid of tinkering"* and became one of their best team members. For many managers, the capacity to learn quickly was more important than current knowledge. P8 referred to this as *"learnability, that ability to climb the curve,"* while P9 looked for whether candidates could *"learn and come up to speed quickly"* and *"continue to learn through their career."* As P12 concluded, the goal is to *"find somebody who is happy to learn"*, a quality that underpins long-term growth and technical advancement.

Process-Oriented Mindset is important, particularly in environments where technical contributions must integrate seamlessly into larger systems. P6 described this as essential *"process thinking"* and emphasized that many technically capable individuals *"can go down a rabbit hole with that skill,"* losing sight of the broader impact. Instead, managers expect candidates to demonstrate *"systems thinking... seeing how does this actually flow through, how does this affect everything else around it, what is the wider scope beyond just this one piece."* This holistic mindset is crucial not just for writing effective code but for designing sustainable workflows. As P6 pointed out, the industry buzz around automation often oversimplifies the real challenge: *"writing a script to do the thing we were doing manually... is not really helping anyone"* if it ignores long-term maintainability. Software Engineering candidates need the ability to *"come at it with a fresh mind on how I can change the process to make it smoother,"* combining critical thinking with technical efficiency. P12 reinforced this for software engineers from a release management perspective, noting the industry expectation for candidates to be *"intentional with releases,"* including an understanding of *"versioning, semantic versioning, bundling."*

RQ1 Our analysis shows that HPE hiring managers prioritize non-technical skills over specific technical proficiencies, as these are seen as harder to teach and more predictive of long-term success. The expected non-technical skills include adaptability, proactivity, dependability, self-direction, passion, collaboration, communication, problem solving, courage, self-awareness, self-confidence, willingness to learn, and a process-oriented mindset. Technical skills are still expected but viewed as easier to develop on the job.

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B. RQ. How are these expected skills reflected in job postings for entry-level Software Engineering-related positions?

The first author of the paper collected seven job postings from different business units from HPE, which are entry-level, require a Computer Science degree, and are related to Software Engineering-related positions: Software Designer I, Software Engineer Firmware I, Systems/Software Engineer I, Cloud Developer I, Product Manager I, Software Engineer Quality Assurance I, and Test Engineer I. The complete text of the job postings is provided in the replication package.

The leveraged GPT-o3-mini OpenAI API version, to prompt the job posting and, by having RAG read the definition of skills that we provided from RQ1, identify which skills the job posting includes. Our prompt script used the role prompting strategy [18] which aims to sweeten the responses of an LLM by leveraging the model's ability to answer as an expert would do. Role prompting gives the message sent to the LLM a prelude characterizing the expert role. The expected outcome is that the LLM answers the prompted question using the context of the presented role.

To inform the retrieval component, we used the skills we found through qualitative analysis of interviews (see Fig.2), including the category name of the skill and the complete description (Sec. IV-A), thereby enhancing the RAG model's contextual alignment with the customized organizational expectations from HPE hiring managers.

We used the prompt: *"You are a SE technical manager with experience with hiring entry-level candidates for jobs, which of the skills – adaptable, proactive, self-directed, dependable, passionate, collaborative, communicative, problem-solver, courageous, self-aware, self-confident, willing to learn, process-oriented mindset, cloud computing, DevOps, operating systems, version control systems, open-source software - does the following job posting need?"* followed by the job posting data. The prompt was formulated for all seven job postings available.

To assess the alignment between job postings and managerial expectations, we manually compared the skills explicitly mentioned by managers during the member-checking survey (see Sec. III-E) with those identified by RAG-based analysis.

As shown in Table IV, we found only partial alignment for non-technical skills. In particular, WILLING TO LEARN was the only skill that consistently matched in all roles where managers expected it and was stated in job postings. This widespread agreement reinforces its centrality in early career expectations and suggests that it is more likely to be clearly communicated by HR teams.

However, several important attributes emphasized by managers, such as PROACTIVE, SELF-DIRECTED, PASSIONATE, and SELF-AWARE, were rarely reflected in the postings. In contrast, some skills, such as CLOUD COMPUTING and being SELF-DIRECTED, appeared in the RAG results, but were not explicitly raised by the managers in the follow-up survey, potentially due to semantic variation or contextual assumptions.

We then used a second prompt *"For every skill you detect in the job posting, provide the exact excerpt or sentence from the job posting where the skill appears and the words or expression from the description of the skill that you used to trigger your classification of that skill. Format your response as a list of skill–evidence pairs."*

We analyzed the RAG-generated summaries for each job posting to identify which skills were emphasized, and where in the job postings the RAG model identified each skill. For the Systems/Software Engineer I role, for instance, the term associated with SELF-DIRECTED was *"independent judgment"*, extracted from the sentence: *"Exercises independent judgment within defined parameters."* For COLLABORATIVE, relevant excerpts included: *"Acts as team member by providing information, analysis, and recommendations in support of team efforts"* and *"Participates as a member of project team..."* These examples illustrate how certain skill indicators may appear implicitly through role descriptions. The complete mapping of skills to text excerpts across all postings is available in our replication package [8].

These results highlight the need for closer collaboration between SE hiring managers and HR teams to ensure that job postings better reflect actual expectations. When essential skills are undercommunicated or ambiguously described, promising candidates may misinterpret role requirements or miss opportunities to tailor their applications. Aligning language and framing across internal expectations and external postings is especially critical in skills-based hiring, where clarity and consistency help reduce mismatches and improve recruitment outcomes.

RQ2 Many of the non-technical skills highly valued by managers, like proactivity, self-awareness, and collaboration, were missing or vaguely described in job postings. Willingness to learn was consistently aligned across expectations and postings. This gap may mislead applicants and hinder skills-based hiring.

V. DISCUSSION

We discuss our findings, threats to validity, and implications.

A. Discussion of Findings

Early-career software engineers need the non-technical skills. Prior work has repeatedly shown that non-technical skills are not only essential for team productivity and project success but also among the most common gaps identified in graduates entering the workforce [1]. Our study reinforces the view that software engineering is a fundamentally socio-technical discipline [19], where success at the entry-level depends on a complex blend of technical fluency and non-technical competencies. The emphasis hiring managers placed on skills like adaptability, communication, and collaboration (see Fig. 2) aligns with a SE research highlighting the limitations of a purely technical definition of engineering readiness [1] and the Computer Science curricula that includes technical competencies but also professional dispositions [17]. Our results support Garcia et al.'s study [1] that communication, collaboration, and willingness to learn are especially valued across global computing employers, and echo previous research

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TABLE IV: Comparison of Expected and Stated Skills Across Job Postings

Job Postings	Skills expected by managers for each role	Job postings analyzed with RAG
Systems/SW Engineer I	Adaptable(3), **Collaborative**(3), **Communicative**(3), Courageous, **Dependable**(2), Passionate, Proactive(3), **Problem-Solver**(3), Process-Oriented Mindset, Self-Directed(3), **Willing to Learn**(4), Self-Aware(2), Self-Confident, Cloud, DevOps(2), **OS**(3), OSS, VCS(4).	**Collaborative**, **Communicative**, **Dependable**, **Problem-Solver**, **Willing to Learn**, **OS**.
SW Engineer Firmware I	**Collaborative**(1), Passionate, Proactive, Self-Confident, Self-Directed, **Willing to Learn**(1), OS, OSS, VCS.	**Collaborative**, Communicative, Dependable, Problem-Solver, Process-Oriented Mindset, **Willing to Learn**.
Cloud Developer I	**Collaborative**(1), **Communicative**(1), Dependable, **Proactive**(1), **Problem Solver**(1), Self-Aware, **Willing to Learn**, **Cloud**(2), **DevOps**(2), OSS, VCS(3),	Adaptable, **Collaborative**, **Communicative**, **Proactive**, **Problem-Solver**, Self-Directed, **Willing to Learn**, **Cloud**, **DevOps**
Product Manager I	Adaptable, **Communicative**(1), Proactive, **Problem Solver**(1), Self-Confident, **Willing to Learn**(1), Cloud, OSS	Collaborative, **Communicative**, **Problem-Solver**, Process-Oriented Mindset, Self-directed, **Willing to Learn**.
Test Engineer I	**Adaptable**(1), **Collaborative**(1), Proactive, **Process-Oriented Mindset**(1), **Willing to Learn**(1), Cloud, DevOps, OS, VCS.	**Adaptable**, **Collaborative**, Communicative, Dependable, Problem-solver, **Process-Oriented Mindset**, **Willing to Learn**.
SW Designer I	**Adaptable**(2), **Collaborative** (2), **Communicative**(1), Proactive, **Problem Solver**, **Process-Oriented Mindset**(1), **Willing to Learn**(2), DevOps, OS(2), VCS(2), OSS.	**Adaptable**, **Collaborative**, **Communicative**, Dependable, **Problem-Solver**, **Process-Oriented Mindset**, Self-Directed, **Willing to Learn**.
SW Quality Assurance I	**Dependable**(1), Proactive, **Problem-solver**(1), Self-Aware, Self-Directed, **Willing to Learn**(1), Cloud, DevOps, OS, VCS.	Collaborative, Communicative, **Dependable**, **Problem-Solver**, Process-Oriented Mindset, **Willing to Learn**, OS, VCS.

Note: Bolded skills indicate a match between those expected by at least one of the five managers (from the member-checking survey) and those identified in the job postings via RAG analysis using qualitative results from RQ1. Numbers in parentheses indicate how many managers expected each skill for the corresponding position. OS = Operational systems, Cloud = Cloud Computing, VCS = Version control systems, OSS = Open Source Software.

[3, 20] that CS education needs to not prioritize programming proficiency over interpersonal or process-oriented learning outcomes.

Invisible skills and measurable ones. Notably, the skills that managers in our study expect, such as self-direction, courage, and process thinking (see Fig. 2), are not always clearly defined in job postings. These "invisible" skills are difficult to assess via resumes or transcripts, but often become decisive during interviews and early performance evaluations. For example, managers described initiative and willingness to learn as more predictive of early-career success than familiarity with any specific technology. This resonates with Assyne et al.'s [21] call for a unified competence framework in SE that accounts for both behavioral and contextual dimensions of developer effectiveness. Interestingly, while technical tools like Git, CI/CD, and cloud computing were still expected, and can be "measured" through GitHub contributions and certificates, they were treated by managers as teachable and secondary. The managers' emphasis on process orientation and systems thinking also extends the SE skills conversation into territory traditionally underexplored in early-career hiring.

If you need that skill, why is it not in the posting? Our results suggest a divergence between the skills that managers find valuable and those articulated in the job post. This finding echoes Wang et al.'s [22] study on job ads for requirements engineers, where relevant domain skills were poorly communicated in postings, contributing to misaligned candidate expectations. Our RQ2 analysis of job postings revealed a structural challenge: even when organizations value non-technical skills, they may lack the language or standard frameworks to consistently express them in recruiting materials. A job posting can be viewed as a communication of both the activities and intentions associated with a new hire's role. When technical skills are framed as specific actions to perform, they may narrow the candidate's interpretation of the role. In contrast, describing skills in terms of outcome intentions, as highlighting the desired results or impact, can foster a deeper understanding of the position's goals [23].

B. Threats to Validity and Limitations

Certain threats to validity might have impacted the validity of your study. In this section, we describe them according to the model of Wohlin et al. [24].

External Validity. The results are valid for the studied suborganization within HPE and may not generalize to other companies, industries, or geographic contexts. Although the participant pool included 12 technical managers with diverse roles, tenures, and genders, enhancing the richness of our insights, further studies are needed to validate whether these expectations hold across other hiring environments.

Internal Validity. Interviews followed a semi-structured protocol and were supported by member checking, but responses may reflect recall bias or emphasize particularly memorable candidates. By encouraging open-ended answers, we captured what managers found most meaningful. For job posting analysis, we used RAG with a fixed set of skills derived from interviews, ensuring the evaluation was grounded in manager expectations rather than shaped by the postings themselves. RAG may hallucinate but we did not observe any case in our sample.

Construct Validity. We used open-ended questions and inductive open coding to capture the skills expected of entry-level SE candidates. Coding was done collaboratively, refined through consensus meetings, and validated by five (out the 12) participants through member checking. To tailor skill detection to HPE, we used RAG to incorporate context from our qualitative coding. Still, some nuances in how skills were expressed may not have been fully captured. In future studies, we aim to increase our sample of interviews and investigate the findings across more job postings.

Conclusion Validity. As with most qualitative studies, our findings are interpretive and not statistically generalizable. We applied multiple validation strategies, as collaborative coding, member checking, and peer review—to support credibility

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[25]. Although we do not claim full saturation, our sample was sufficient to uncover a consistent and comprehensive account of expected skills, as defined by Bernard [9]. A notable limitation is the non-deterministic nature of LLMs used in our RAG approach. Despite fixing model settings, repeated runs may yield slightly different outputs. We mitigated this by manually reviewing RAG results, but future work should explore strategies to improve reproducibility, as multiple inference rounds or snapshot comparisons.

C. Implications for Practice

Our findings will be shared with HPE managers and HR to support better alignment of job postings with actual expectations. More broadly, they highlight the need for software organizations to reevaluate how entry-level skills are assessed, communicated, and supported in practice.

Refactor Job Postings on the Non-Technical Skill Requirements. Our findings show that even widely expected technical competencies, such as cloud computing, DevOps, or version control, were not clearly reflected in job postings. In several cases, technical skills were described in the job postings through vague actions (e.g., "codes and debugs") rather than named domains, making it difficult for candidates to recognize the intended requirements. This lack of clarity also affected non-technical skills, many of which—like proactivity, self-direction, and passion—were emphasized by managers but rarely surfaced in the postings. Only one skill, WILLING TO LEARN, was consistently represented across both expectations and postings, suggesting that more universally understood or frequently used terms are more likely to be included. These mismatches can mislead applicants, distort hiring pipelines, and contribute to missed opportunities for both organizations and candidates. To address this, software organizations should strengthen collaboration between hiring managers and HR teams to ensure that postings accurately reflect role-specific expectations. Job descriptions should move beyond generic task lists and embed clear, behaviorally-anchored skill indicators. Making both technical and non-technical skills more explicit, using terminology consistent with team expectations can help reduce ambiguity, attract better-fit applicants, and improve outcomes in skills-based hiring.

Align Internal Language Between Technical Managers and HR. RQ2 results (see Table IV) show a communication breakdown between what hiring managers value and what appears in job postings. One practical outcome is to co-create a shared vocabulary or template that helps HR and technical leads express expectations consistently. This reduces the risk of misinterpretation and ensures that key skills (e.g. adaptable, self-aware, passionate) are not assumed to be "common sense."

VI. RELATED WORK

Prior research has consistently highlighted the gap between academic preparation and industry expectations for entry-level computing roles, particularly regarding non-technical skills. Garcia et al. [1] conducted a study that revealed misalignments in how professional dispositions, such as communication, collaboration, and lifelong learning, are emphasized in academia versus industry. While offering a broad perspective, their work does not examine how these expectations are articulated in job postings or influence company-level hiring.

Other studies have investigated these gaps in more focused contexts. Akdur et al. [3, 26] surveyed software professionals in the embedded systems domain, reporting that graduates often lack crucial non-technical skills such as communication, teamwork, and adaptability. Similarly, Christensen and Paasivaara [20] demonstrated that distributed development courses can strengthen students' soft skills, though these remain underdeveloped even after technical training. Assyne et al. [21] proposed a comprehensive competence framework for software roles, highlighting both hard and soft skills across project lifecycles, though their focus remained on modeling and assessment rather than hiring practices. Wang et al. [22] further noted inconsistencies between job ads and formal role definitions, such as those in SWEBOK, showing that postings often fail to capture the full range of expected competencies.

While these studies underscore the importance of non-technical competencies and curricular reform, they largely overlook how such expectations are operationalized in hiring artifacts. Our study addresses this gap by comparing the skills expected by hiring managers with those expressed in job postings within a global company.

Recent work has also explored the use of AI and NLP techniques for skill extraction and classification from job-related text. Zhang et al. [27] introduced SkillSpan, a dataset of annotated job postings that used domain-specific adaptation for skill extraction. Other efforts have leveraged machine learning [28–30], transformers [31], and fine-tuned LLMs such as SkillScope [32] and SkillGPT [33] to extract, categorize, and standardize skills across technical domains. These tools have shown that LLMs can reduce preprocessing overhead and improve the precision of skill identification.

Our study fills this gap by directly comparing what hiring managers expect with how those expectations are communicated in job postings. We focus on both technical and non-technical skills, using open coding from interviews and applying RAG to assess alignment. This fine-grained, context-aware analysis advances state-of-the-art by identifying mismatches that may affect early-career hiring in SE.

VII. CONCLUSION

This study examined the disconnect between the skills hiring managers expect and those conveyed in job postings for entry-level SE roles. Based on interviews with 12 HPE managers and analysis of seven job postings, we identified 13 non-technical and five technical skills prioritized by hiring managers. Our analysis revealed that many of these skills are often underrepresented or vaguely stated in job descriptions.

Future work can expand this research across different organizations to refine strategies for aligning candidate preparation with industry needs. By deepening our understanding of how skills are communicated and interpreted, future studies can provide more actionable guidance for aspiring software engineers navigating the early-career hiring landscape.

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